Haloalkanes

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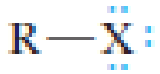
Haloalkanes

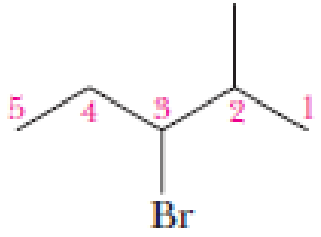
- A haloalkane, commonly known as an alkyl halide, is a compound that contains a halogen atom covalently bonded to an sp^3 hybridised carbon atom.
- Given the symbol RX, where X can be F, Cl, Br, or I.



A haloalkane (An alkyl halide)

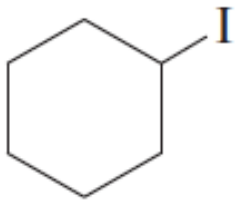
IUPAC NAMES

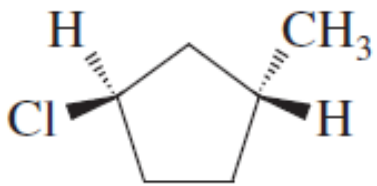
- Locate and name the parent chain containing the largest number of carbon atoms.
- Number the parent chain from the direction that gives the substituent encountered first the lowest number.
- Show halogen substituents by the prefixes *fluoro-*, *chloro-*, *iodo-*, and *bromo-*, and list them in alphabetical order along with other substituents.



3-Bromo-2-methylpentane

	$CH_3CH_2—F$	$CH_2(Cl)CH_2CH_2CH_3$	$CH_3—CH(Br)—CH_3$
IUPAC name:	fluoroethane	1-chlorobutane	2-bromopropane





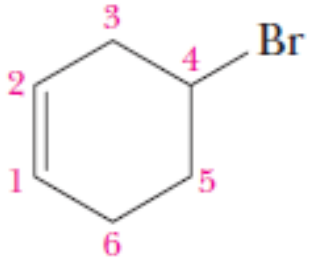
IUPAC name: iodocyclohexane *trans*-1-chloro-3-methylcyclopentane

	$CH_3CH_2—CH(CH_2—I)—CH_2CH_3$	$CH_3CH_2CH_2—CH(CH_2CH_2—F)—CH_2CH_2CH_3$
IUPAC name:	3-(iodomethyl)pentane	4-(2-fluoroethyl)heptane

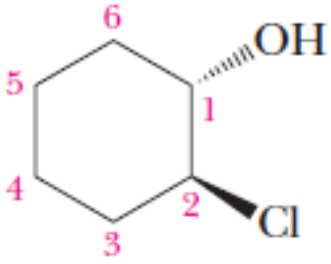
For halo-alkyl substituents, the carbon bonded to the parent chain is numbered as carbon 1. Use a number to indicate position of halogen

	CH_2Cl_2	$CHCl_3$	CCl_4
IUPAC name:	dichloromethane	trichloromethane	tetrachloromethane
common name:	methylene chloride	chloroform	carbon tetrachloride

- In ***haloalkenes***, the location of the double bond determines the numbering of the parent hydrocarbon. In molecules containing functional groups designated by a suffix (for example, *ol*, *ene*, *oic acid*), the location of the functional group indicated by the suffix determines the numbering.



4-Bromocyclohexene



trans-2-Chlorocyclohexanol

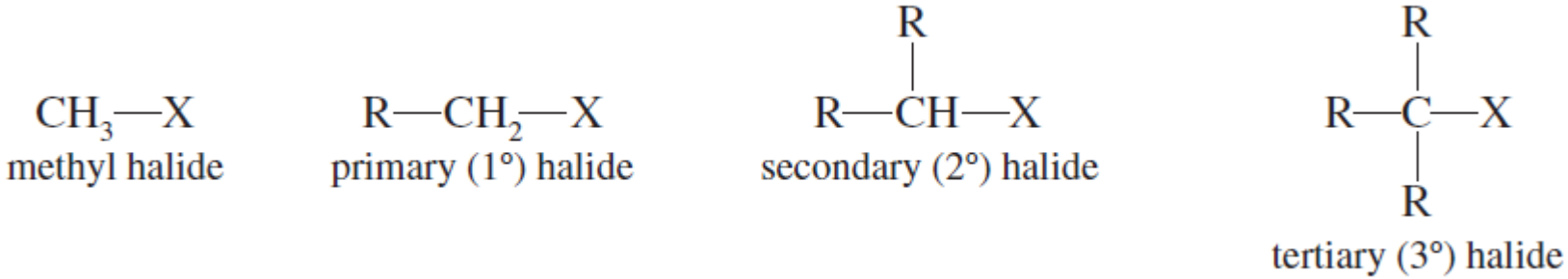
Practice exercise 1: Draw the condensed structural formulae of the following compounds:

3-iodo-2-methylpentane

cis-1-chloro-3-(chloromethyl)cyclohexane

Classification of Haloalkanes

- Alkyl halides are classified according to the nature of the carbon atom which is bonded to the halogen.
- If the carbon is bonded to one other carbon atom, it is a primary (1°) alkyl halide.
- If the carbon is bonded to two other carbon atoms, it is a secondary (2°) alkyl halide, and if the halogen-bearing carbon is bonded to three other carbons, it is a tertiary (3°) alkyl halide.



Where *R* is any carbon-containing group.

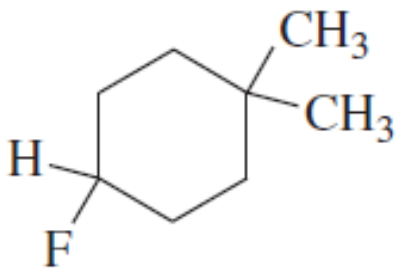
Practice exercise 2: For each of the following compounds, give the IUPAC name and classify the compound as a methyl, primary, secondary, or tertiary alkyl halide.

(a) $(CH_3)_2CHCH_2Cl$

(b) $(CH_3)_3CBr$

(c) $CH_3—CH(CH_2CH_3)—CH_2Cl$

(d)



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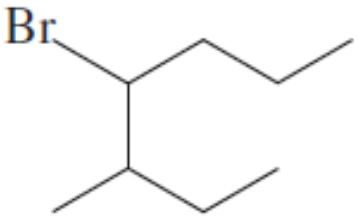

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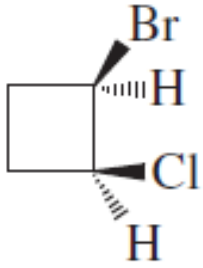


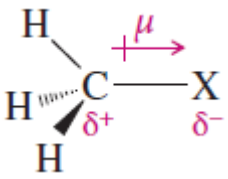
Uses of Haloalkanes

- Solvents – alkyl halides are commonly used in industrial and household solvents for example in cleaning agents, paint removers.
- Reagents – many alkyl halides are used as starting materials for the synthesis of many other complex substances.
- Anaesthetics – ethyl chloride is used a topical anaesthetic for minor procedures. The mixed alkyl halide $CF_3CHClBr$, which goes by the common name Halothane is used as an anaesthetic in surgical procedures.
- Refrigerants – chlorofluorocarbons (also known as Freons) are fluorinated haloalkanes commonly used as refrigerant gas. Freon-22 or $CHClF_2$ is used in refrigerators and automotive air-conditioners.

$CHCl_3$	$CHClF_2$	$CCl_3—CH_3$	$CF_3—CHClBr$
solvent	refrigerant	cleaning fluid	nonflammable anesthetic
chloroform	Freon-22®	1,1,1-trichloroethane	Halothane

Structure of Haloalkanes

- In an alkyl halide, the halogen atom is bonded to an sp^3 carbon atom.
- The halogen is more electronegative than the carbon, thus the C—X bond is polarized.



- The electronegativities of the halogens increase in the order:

	I	<	Br	<	Cl	<	F
electronegativity:	2.7		3.0		3.2		4.0

- The C—X bond lengths increase as the halogen atoms become larger.

	C—F	<	C—Cl	<	C—Br	<	C—I
bond length:	1.38 Å		1.78 Å		1.94 Å		2.14 Å

Physical Properties of Haloalkanes

1) <u>Boiling Points:</u>

- There are two types of intermolecular forces that affect the boiling points of alkyl halides: dispersion forces (London forces) and dipole-dipole forces.
- The larger the surface area of the molecule, the stronger the dispersion forces, the higher the boiling point.
- Surface areas increase with molecular weight.
- Surface areas decrease with increased branching of the molecule.

ethyl fluoride, bp −38 °C
smallest surface area, lowest bp

ethyl chloride, bp 12 °C

ethyl bromide, bp 38 °C

ethyl iodide, bp 72 °C
largest surface area, highest bp

The heavier the halogen atoms are, the greater the surface area for interaction, and the higher the boiling point.
Thus the boiling points of ethyl halides increase in the order: F<Cl<Br<I

2) <u>Densities</u>

- Like their boiling points, their densities also follow predictable trends.
- Alkyl fluorides are less dense than water.
- Alkyl chlorides that contain one chlorine atom are also less dense than water.
- Alkyl chlorides that contain two or more chlorine atoms are more dense than water.
- Alkyl bromides and alkyl iodides are more dense than water.

Practice exercise 3: For each pair of compounds, indicate which compound is likely to have the higher boiling point, and explain why.

a) Isopropyl bromide (2-bromopropane) and *n*-butyl bromide (bromobutane)

b) Isopropyl chloride (2-chloropropane) and *tert*-butyl bromide (2-bromo-2-methylpropane)

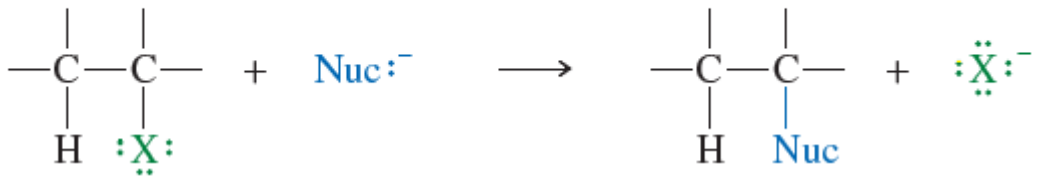
c) 1-bromobutane and 1-chlorobutane

Reactions of Haloalkanes

- Alkyl halides can easily be converted to other functional groups.
- The halogen can leave, carrying with it both electrons from the C—X bond, leaving as a halide ion.
- We say that halides are ***good leaving groups***.
- When another atom/ group of atoms replaces the halogen, the reaction is a substitution.

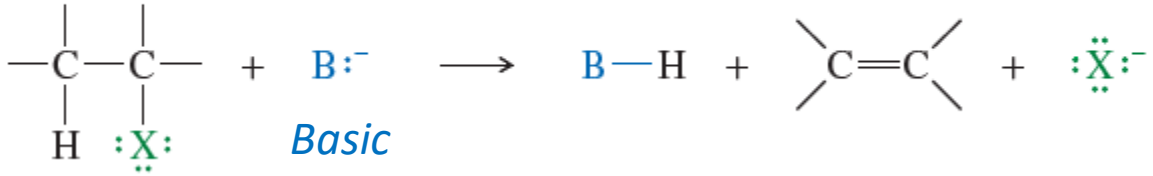
- In a *nucleophilic substitution*, a nucleophile replaces a leaving group from a carbon atom, using its lone pair of electrons to form a new bond with the carbon atom.

Nucleophilic substitution



- When the halide ion leaves with another atom or ion, a new *pi* bond forms, and the result is an *elimination*. In many eliminations, a molecule of H—X is lost from the alkyl halide to form an alkene (known as a dehydrohalogenation).

Elimination



Most nucleophiles act as bases